

Section 8.3 Exercises

Directions: For exercises 1 - 5, convert the given polar coordinates to Cartesian coordinates.

1. $\left(7, \frac{7\pi}{6}\right)$

2. $(5, \pi)$

3. $\left(6, -\frac{\pi}{4}\right)$

4. $\left(-3, \frac{\pi}{6}\right)$

5. $\left(4, \frac{7\pi}{4}\right)$

Directions: For exercises 6 - 10, convert the given Cartesian coordinates to polar coordinates with $r > 0$, $0 \leq \theta < 2\pi$. For non-unit circle angles, round answers to three decimal places.

6. $(4, 2)$

7. $(-4, 6)$

8. $(3, -5)$

9. $(-10, -13)$

10. $(8, 8)$

Directions: For exercises 11 - 22, convert the given Cartesian equation to a polar equation.

11. $x = 3$

12. $y = 4$

13. $y = 4x^2$

14. $y = 2x^4$

15. $x^2 + y^2 = 4y$

16. $x^2 + y^2 = 3x$

17. $x^2 - y^2 = x$

18. $x^2 - y^2 = 3y$

19. $x^2 + y^2 = 9$

20. $x^2 = 9y$

21. $y^2 = 9x$

22. $9xy = 1$

Directions: For exercises 23 - 34, convert the given polar equation to a Cartesian equation.

23. $r = 3\sin \theta$

24. $r = 4\cos \theta$

25. $r = \frac{4}{\sin \theta + 7\cos \theta}$

26. $r = \frac{6}{\cos \theta + 3\sin \theta}$

27. $r = 2\sec \theta$

28. $r = 3\csc \theta$

29. $r = \sqrt{r\cos \theta + 2}$

30. $r^2 = 4\sec \theta \csc \theta$

31. $r = 4$

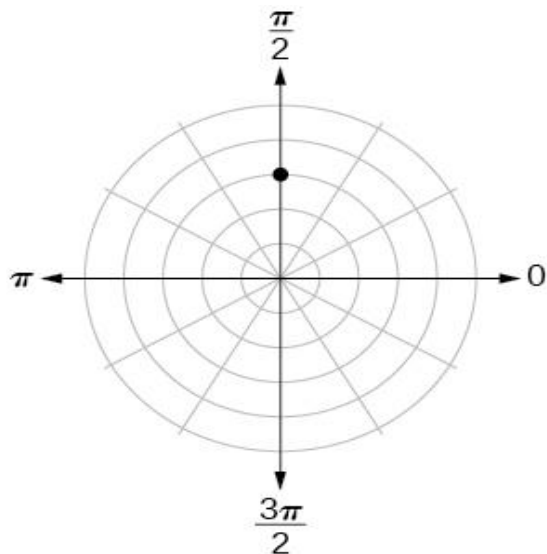
32. $r^2 = 4$

33. $r = \frac{1}{4\cos \theta - 3\sin \theta}$

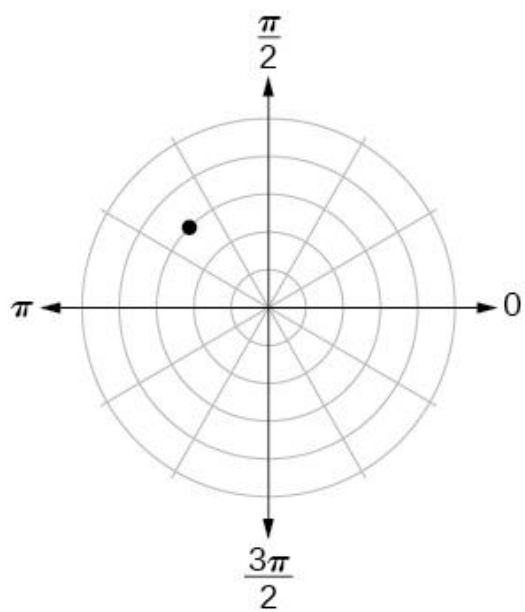
34. $r = \frac{3}{\cos \theta - 5\sin \theta}$

Directions: For exercises 35 - 39, find the polar coordinates of the point.

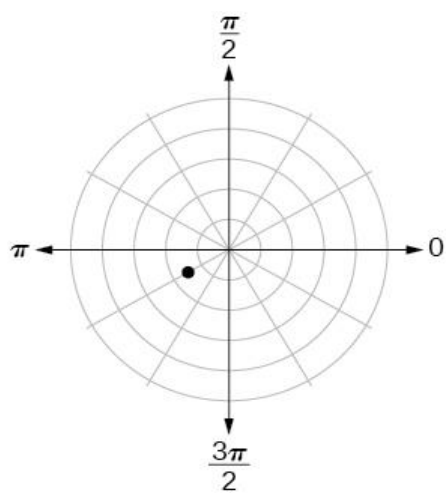
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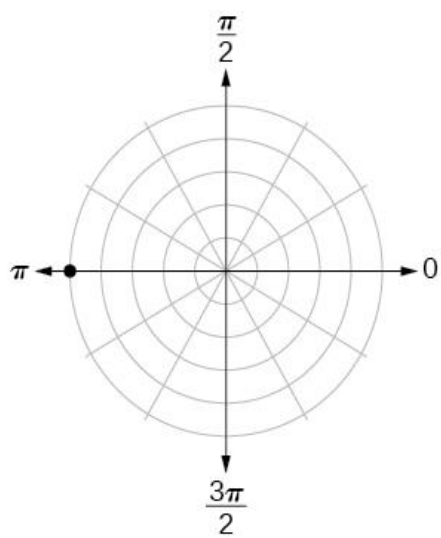
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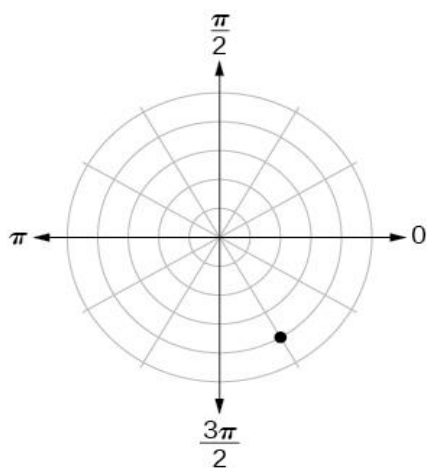
37.



38.



39.



Directions: For exercises 40 - 49, plot the points.

40. $\left(-2, \frac{\pi}{3}\right)$

41. $\left(-1, -\frac{\pi}{2}\right)$

42. $\left(3.5, \frac{7\pi}{4}\right)$

43. $\left(-4, \frac{\pi}{3}\right)$

44. $\left(5, \frac{\pi}{2}\right)$

45. $\left(4, \frac{-5\pi}{4}\right)$

46. $\left(3, \frac{5\pi}{6}\right)$

47. $\left(-1.5, \frac{7\pi}{6}\right)$

48. $\left(-2, \frac{\pi}{4}\right)$

49. $\left(1, \frac{3\pi}{2}\right)$